Sungsoo Ahn

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Education

Integrated M.S.–Ph.D. Program in Electrical Engineering, KAIST, Daejeon, South Korea, Advisor: Jinwoo Shin, Mar. 2015 – Mar. 2021.

B.S. in Electrical Engineering, KAIST, Daejeon, South Korea, Mar. 2011 – Mar. 2015.

Employment

Assistant Professor, Graduate School of AI, KAIST, Daejeon, South Korea, Jan. 2025 – Present.

Assistant Professor, Graduate School of AI and Department of Computer Science and Engineering, POSTECH, Pohang, South Korea, Dec. 2021 – Jan. 2025.

Postdoctoral Research Associate, Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), Abu Dhabi, United Arab Emirates, Supervisors: Le Song and Eric Xing, Mar. 2021 – Nov. 2021.

Research Intern, Amazon Cambridge Development Center, Cambridge, United Kingdom, Supervisor: Zhenwen Dai, Jun. 2018 – Aug. 2018.

Visiting Student, University of Cambridge, Cambridge, United Kingdom, Host: Adrian Weller, Mar. 2018 – May 2018.

Research Intern, Los Alamos National Laboratory, Los Alamos, New Mexico, United States, Host: Michael Chertkov, Jun. 2017 – Aug. 2017.

Research Intern, Los Alamos National Laboratory, Los Alamos, New Mexico, United States, Host: Michael Chertkov, Jun. 2016 – Aug. 2016.

Software Developer Intern, Penta Security Systems, Seoul, South Korea, Jan. 2014 – Feb. 2014.

Publications

*Equal contribution. †Equal corresponding authors.

Conference Papers

1. Yinhua Piao, Hyomin Kim, Seonghwan Kim, Yunhak Oh, Junhyeok Jeon, Sang-Yeon Hwang, Jaechang Lim, Woo Youn Kim, Chanyoung Park, and **Sungsoo Ahn**, Learning Adaptive Perturbation-Conditioned Contexts for Robust Transcriptional Response Prediction, in *ICML*, 2026. [arXiv]

2. Kiyoung Seong, **Sungsoo Ahn**, Sehui Han, and Changyoung Park, Multimodal Crystal Flow: Any-to-Any Modality Generation for Unified Crystal Modeling, in *ICML*, 2026. [arXiv]
3. Dongyeop Woo, Marta Skreta, Seonghyun Park, Kirill Neklyudov†, and **Sungsoo Ahn**†, Riemannian MeanFlow, in *ICML*, 2026. [arXiv] [code]
4. Seongsu Kim, Chanhui Lee, Yoonho Kim, Seongjun Yun, Honghui Kim, Nayoung Kim, Changyoung Park, Sehui Han, Sungbin Lim†, and **Sungsoo Ahn**†, Machine Learning Hamiltonians are Accurate Energy-Force Predictors, in *ICML*, 2026. [arXiv] [code]
5. Minkyu Kim, Nayoung Kim, Honghui Kim, and **Sungsoo Ahn**, CatFlow: Co-generation of Slab-Adsorbate Systems via Flow Matching, in *ICML*, 2026. [arXiv] [code]
6. Taewon Kim, Jihwan Shin, Hyomin Kim, Youngmok Jung, Jonghoon Lee, Won-Chul Lee, **Sungsoo Ahn**†, and Insu Han†, DNACHUNKER: Learnable Tokenization for DNA Language Models, in *ICML*, 2026. [arXiv] [project]
7. Minsu Kim, Jean-Pierre R. Falet, Oliver Ethan Richardson, Xiaoyin Chen, Moksh Jain, Sungjin Ahn, **Sungsoo Ahn**, and Yoshua Bengio, Latent Veracity Inference for Identifying Errors in Stepwise Reasoning, in *ICLR*, 2026. [paper] [arXiv] [code]
8. Hyunjin Seo*, Taewon Kim*, Sihyun Yu, and **Sungsoo Ahn**, Learning Flexible Forward Trajectories for Masked Molecular Diffusion, in *ICLR*, 2026. [paper] [arXiv] [code] [project]
9. Seonghyun Park, Kiyoung Seong, Soojung Yang, Rafael Gómez-Bombarelli, and **Sungsoo Ahn**, Learning Collective Variables from BioEmu with Time-Lagged Generation, in *ICLR*, 2026. [paper] [arXiv] [code]
10. Dongyeop Woo, Minsu Kim, Minkyu Kim, Kiyoung Seong, and **Sungsoo Ahn**, Energy-based Generator Matching: A Neural Sampler for General State Space, in *NeurIPS*, 2025. [arXiv] [code]
11. Minkyu Kim, Kiyoung Seong, Dongyeop Woo, **Sungsoo Ahn**, and Minsu Kim, On Scalable and Efficient Training of Diffusion Samplers, in *NeurIPS*, 2025. [arXiv] [code]
12. Nayoung Kim, Seongsu Kim, and **Sungsoo Ahn**, Flexible MOF Generation with Torsion-Aware Flow Matching, in *NeurIPS*, 2025. [arXiv] [code]
13. Seongsu Kim, Nayoung Kim, Dongwoo Kim, and **Sungsoo Ahn**, High-order Equivariant Flow Matching for Density Functional Theory Hamiltonian Prediction, in *NeurIPS*, 2025, spotlight presentation. [arXiv] [code]
14. Hyosoon Jang, Yunhui Jang, Sungjae Lee, Jungseul Ok, and **Sungsoo Ahn**, Self-Training Large Language Models with Confident Reasoning, in *Findings of EMNLP*, 2025. [arXiv] [code]
15. Hyomin Kim, Yunhui Jang, and **Sungsoo Ahn**, MT-Mol: Multi Agent System with Tool-based Reasoning for Molecular Optimization, in *Findings of EMNLP*, 2025. [arXiv] [code]
16. Yunhui Jang, Jaehyung Kim, and **Sungsoo Ahn**, Improving Chemical Understanding of LLMs via SMILES Parsing, in *EMNLP*, 2025. [arXiv] [code] [project]

17. Federico Berto, Chuanbo Hua, Junyoung Park, Laurin Luttmann, Yining Ma, Fanchen Bu, Jiarui Wang, Haoran Ye, Minsu Kim, Sanghyeok Choi, Nayeli Gast Zepeda, André Hottung, Jianan Zhou, Jieyi Bi, Yu Hu, Fei Liu, Hyeonah Kim, Jiwoo Son, Haeyeon Kim, Davide Angioni, Wouter Kool, Zhiguang Cao, Qingfu Zhang, Joungho Kim, Jie Zhang, Kijung Shin, Cathy Wu, **Sungsoo Ahn**, Guojie Song, Changhyun Kwon, Kevin Tierney, Lin Xie, and Jinkyoo Park, RL4CO: an Extensive Reinforcement Learning for Combinatorial Optimization Benchmark, in *KDD, Datasets and Benchmarks Track*, 2025. [arXiv] [code]

18. Dongyoon Hahm, Woogyeol Jin, June Suk Choi, **Sungsoo Ahn**, and Kimin Lee, Enhancing LLM Agent Safety via Causal Influence Prompting, in *Findings of ACL*, 2025. [arXiv] [code]

19. Yunhui Jang, Jaehyung Kim, and **Sungsoo Ahn**, Structural Reasoning Improves Molecular Understanding of LLM, in *ACL*, 2025. [arXiv] [code] [project]

20. Seonghwan Seo, Minsu Kim, Tony Shen, Martin Ester, Jinkyoo Park, **Sungsoo Ahn**, and Woo Youn Kim, Generative Flows on Synthetic Pathway for Drug Design, in *ICLR*, 2025. [paper] [arXiv] [code]

21. Nayoung Kim, Seongsu Kim, Minsu Kim, Jinkyoo Park, and **Sungsoo Ahn**, MOFFlow: Flow Matching for Structure Prediction of Metal-Organic Frameworks, in *ICLR*, 2025. [paper] [arXiv] [code]

22. Kiyoung Seong, Seonghyun Park, Seonghwan Kim, Woo Youn Kim, and **Sungsoo Ahn**, Transition Path Sampling with Improved Off-Policy Training of Diffusion Path Samplers, in *ICLR*, 2025. [paper] [arXiv] [code] [project]

23. Taewon Kim*, Hyunjin Seo*, **Sungsoo Ahn**, and Eunho Yang, REBIND: Enhancing Ground-state Molecular Conformation Prediction via Force-Based Graph Rewiring, in *ICLR*, 2025. [paper] [arXiv] [code]

24. Minsu Kim*, Sanghyeok Choi*, Taeyoung Yun, Emmanuel Bengio, Leo Feng, Jarrid Rector-Brooks, **Sungsoo Ahn**, Jinkyoo Park, Nikolay Malkin, and Yoshua Bengio, Adaptive Teachers for Amortized Samplers, in *ICLR*, 2025. [paper] [arXiv] [code]

25. Hyosoon Jang, Yunhui Jang, Minsu Kim, Jinkyoo Park, and **Sungsoo Ahn**, Pessimistic Backward Policy for GFlowNets, in *NeurIPS*, 2024. [paper] [arXiv] [code]

26. Jaeseung Heo, Seungbeom Lee, **Sungsoo Ahn**, and Dongwoo Kim, EPIC: Graph Augmentation with Edit Path Interpolation via Learnable Cost, in *IJCAI*, 2024. [paper] [arXiv]

27. Hyomin Kim, Yunhui Jang, Jaeho Lee, and **Sungsoo Ahn**, Hybrid Neural Representations for Spherical Data, in *ICML*, 2024. [paper] [arXiv] [code]

28. Seongsu Kim and **Sungsoo Ahn**, Gaussian Plane-Wave Neural Operator for Electron Density Estimation, in *ICML*, 2024. [paper] [arXiv] [code]

29. Nayeong Kim, Juwon Kang, **Sungsoo Ahn**, Jungseul Ok, and Suha Kwak, Improving Robustness to Multiple Spurious Correlations by Multi-Objective Optimization, in *ICML*, 2024. [paper] [arXiv] [code]

30. Hyeonah Kim, Minsu Kim, **Sungsoo Ahn**, and Jinkyoo Park, Symmetric Replay Training:

Enhancing Sample Efficiency in Deep Reinforcement Learning for Combinatorial Optimization, in *ICML*, 2024. [arXiv] [code]

31. Fanchen Bu, Hyeonsoo Jo, Soo Yong Lee, **Sungsoo Ahn**, and Kijung Shin, Tackling Prevalent Conditions in Unsupervised Combinatorial Optimization: Cardinality, Minimum, Covering, and More, in *ICML*, 2024. [paper] [arXiv] [code]
32. Youngsik Yoon, Gangbok Lee, **Sungsoo Ahn**, and Jungseul Ok, Breadth-First Exploration on Adaptive Grid for Reinforcement Learning, in *ICML*, 2024. [paper] [code] [project]
33. Hyosoon Jang, Minsu Kim, and **Sungsoo Ahn**, Learning Energy Decompositions for Partial Inference in GFlowNets, in *ICLR*, 2024, oral presentation (86 of 7,304 submissions, 1.2%). [paper] [arXiv] [code]
34. Yunhui Jang, Seul Lee, and **Sungsoo Ahn**, A Simple and Scalable Representation for Graph Generation, in *ICLR*, 2024. [paper] [arXiv] [code] [project]
35. Yunhui Jang, Dongwoo Kim, and **Sungsoo Ahn**, Graph Generation with K^2-trees, in *ICLR*, 2024. [paper] [arXiv] [code] [project]
36. Minsu Kim, Taeyoung Yun, Emmanuel Bengio, Dinghuai Zhang, Yoshua Bengio, **Sungsoo Ahn**, and Jinkyoo Park, Local Search GFlowNets, in *ICLR*, 2024, spotlight presentation (366 of 7,304 submissions, 5.0%). [paper] [arXiv] [code]
37. Hyuna Cho, Minjae Jeong, Sooyeon Jeon, **Sungsoo Ahn**, and Won Hwa Kim, Multi-resolution Spectral Coherence for Graph Generation with Score-based Diffusion, in *NeurIPS*, 2023. [paper]
38. Hyosoon Jang, Seonghyun Park, Sangwoo Mo, and **Sungsoo Ahn**, Diffusion Probabilistic Models for Structured Node Classification, in *NeurIPS*, 2023. [paper] [arXiv] [code]
39. Minsu Kim, Federico Berto, **Sungsoo Ahn**, and Jinkyoo Park, Bootstrapped Training of Score-Conditioned Generator for Offline Design of Biological Sequences, in *NeurIPS*, 2023. [paper] [arXiv] [code]
40. Sungbin Shin, Yohan Jo, **Sungsoo Ahn**, and Namhoon Lee, A Closer Look at the Intervention Procedure of Concept Bottleneck Models, in *ICML*, 2023. [paper] [arXiv] [code]
41. Junsu Kim, Younggyo Seo, **Sungsoo Ahn**, Kyunghwan Son, and Jinwoo Shin, Imitating Graph-Based Planning with Goal-Conditioned Policies, in *ICLR*, 2023. [paper] [arXiv] [code]
42. Nayeong Kim, Sehyun Hwang, **Sungsoo Ahn**, Jaesik Park, and Suha Kwak, Learning Debiased Classifier with Biased Committee, in *NeurIPS*, 2022. [paper] [arXiv] [code]
43. Kyunghwan Son, Junsu Kim, **Sungsoo Ahn**, Roben Delos Reyes, Yung Yi, and Jinwoo Shin, Disentangling Sources of Risk for Distributional Multi-Agent Reinforcement Learning, in *ICML*, 2022. [paper]
44. Jaehyung Kim, Dongyeop Kang, **Sungsoo Ahn**, and Jinwoo Shin, What Makes Better Augmentation Strategies? Augment Difficult but Not Too Different, in *ICLR*, 2022. [paper] [code]

45. **Sungsoo Ahn**, Binghong Chen, Tianzhe Wang, and Le Song, Spanning Tree-based Graph Generation for Molecules, in *ICLR*, 2022, spotlight presentation (174 of 3,422 submissions, 5.1%). [paper]
46. Sihyun Yu, **Sungsoo Ahn**, Le Song, and Jinwoo Shin, RoMA: Robust Model Adaptation for Offline Model-based Optimization, in *NeurIPS*, 2021. [arXiv] [code]
47. Hankook Lee, **Sungsoo Ahn**, Seung-Woo Seo, You Young Song, Eunho Yang, Sung-Ju Hwang, and Jinwoo Shin, RetCL: A Selection-based Approach for Retrosynthesis via Contrastive Learning, in *IJCAI*, 2021. [paper] [arXiv] [code]
48. Junsu Kim, **Sungsoo Ahn**, Hankook Lee, and Jinwoo Shin, Self-Improved Retrosynthetic Planning, in *ICML*, 2021. [paper] [arXiv] [code]
49. Jaeho Lee, Sejun Park, Sangwoo Mo, **Sungsoo Ahn**, and Jinwoo Shin, Layer-adaptive sparsity for the Magnitude-based Pruning, in *ICLR*, 2021. [paper] [arXiv] [code]
50. Junhyun Nam, Hyuntak Cha, **Sungsoo Ahn**, Jaeho Lee, and Jinwoo Shin, Learning from Failure: Training Debiased Classifier from Biased Classifier, in *NeurIPS*, 2020. [paper] [arXiv] [code]
51. **Sungsoo Ahn**, Junsu Kim, Hankook Lee, and Jinwoo Shin, Guiding Deep Molecular Optimization with Genetic Exploration, in *NeurIPS*, 2020. [paper] [arXiv] [code]
52. **Sungsoo Ahn**, Younggyo Seo, and Jinwoo Shin, Learning What to Defer for Maximum Independent Sets, in *ICML*, 2020. [paper] [arXiv] [code]
53. **Sungsoo Ahn**, Shell Xu Hu, Andreas Damianou, Neil D. Lawrence, and Zhenwen Dai, Variational Information Distillation for Knowledge Transfer, in *CVPR*, 2019. [paper] [arXiv] [code]
54. **Sungsoo Ahn**, Michael Chertkov, Adrian Weller, and Jinwoo Shin, Bucket Renormalization for Approximate Inference, in *ICML*, 2018. [paper] [arXiv] [code]
55. **Sungsoo Ahn**, Michael Chertkov, Jinwoo Shin, and Adrian Weller, Gauged Mini-Bucket Elimination for Approximate Inference, in *AISTATS*, 2018. [paper] [arXiv] [code]
56. **Sungsoo Ahn**, Michael Chertkov, and Jinwoo Shin, Gauging Variational Inference, in *NeurIPS*, 2017. [paper] [arXiv]
57. **Sungsoo Ahn**, Michael Chertkov, and Jinwoo Shin, Synthesis of MCMC and Belief Propagation, in *NeurIPS*, 2016, oral presentation (46 of 2,500 submissions, 1.8%). [paper] [arXiv]
58. **Sungsoo Ahn**, Sejun Park, Michael Chertkov, and Jinwoo Shin, Minimum Weight Perfect Matching via Blossom Belief Propagation, in *NeurIPS*, 2015, spotlight presentation (82 of 1,838 submissions, 4.5%). [paper] [arXiv]

Journal Articles

1. Nayoung Kim, Minsu Kim, **Sungsoo Ahn**, and Jinkyoo Park, Decoupled Sequence and Structure Generation for Realistic Antibody Design, *Transactions of Machine Learning Research (TMLR)*, 2024. [paper] [arXiv] [code]

2. Seonghyun Park*, Narae Ryu*, Gahee Kim, Dongyeop Woo, Se-Young Yun†, and **Sungsoo Ahn**†, Non-backtracking Graph Neural Networks, *Transactions of Machine Learning Research (TMLR)*, 2024. [paper] [arXiv] [code]

3. Seojin Kim, Jaehyun Nam, Junsu Kim, Hankook Lee, **Sungsoo Ahn**, and Jinwoo Shin, Holistic Molecular Representation Learning via Multi-view Fragmentation, *Transactions of Machine Learning Research (TMLR)*, 2024. [paper] [code]

4. **Sungsoo Ahn**, Michael Chertkov, Adrian Weller, and Jinwoo Shin, Bucket Renormalization for Approximate Inference, *Journal of Statistical Mechanics: Theory and Experiment*, 2019(12), 124015, 2019. [paper] [arXiv] [code]

5. **Sungsoo Ahn**, Michael Chertkov, and Jinwoo Shin, Gauging Variational Inference, *Journal of Statistical Mechanics: Theory and Experiment*, 2019(12), 124022, 2019. [paper] [arXiv]

6. **Sungsoo Ahn**, Michael Chertkov, Andrew E. Gelfand, Sejun Park, and Jinwoo Shin, Maximum Weight Matching using Odd-sized Cycles: Max-Product Belief Propagation and Half-Integrality, *IEEE Transactions on Information Theory*, 64(3), 1471–1480, 2018. [paper]

Preprints

1. Nayoung Kim, Kiyoung Seong, and **Sungsoo Ahn**, Packora: Systematic Design for Generative Molecular Crystal Structure Prediction, *arXiv*, 2026. [arXiv]

2. Kiyoung Seong, Nayoung Kim, and **Sungsoo Ahn**, Discovering Crystal Structure Prediction Algorithms with an AI Co-Scientist, *arXiv*, 2026. [arXiv] [MaskGXT code] [HACO code]

3. Yoonho Kim, Seongsu Kim, **Sungsoo Ahn**, and Honghui Kim, MADField: Multi-fidelity Amortized Density Field for Adsorption in Nanoporous Materials, *arXiv*, 2026. [arXiv]

4. Taewon Kim, Hyosoon Jang, Hyunjin Seo, Seonghwan Seo, Hyeongwoo Kim, Wonho Zhung, Mingyeong Shin, Wooyoun Kim, and **Sungsoo Ahn**, Atom-level Protein Representation Learning Improves Protein Structure Prediction, *arXiv*, 2026. [arXiv] [project]

5. Hyunjin Seo, Hongjoon Ahn, Jimin Park, Sungjun Han, Gyubok Lee, Soojung Yang, Joseph S Brown, Leo Chen, Gina El Nesr, Feyisayo Eweje, Sarah Gurev, Hyejin Lee, Cheng-Hao Liu, Junlang Liu, Zhihui Qi, Jason Yang, Gyu Rie Lee, **Sungsoo Ahn**, Sangwon Jung, and Jamin Shin, VibeProteinBench: An Evaluation Benchmark for Language-interfaced Vibe Protein Design, *arXiv*, 2026. [arXiv] [dataset]

6. Hyomin Kim, Sang-Yeon Hwang, Jaechang Lim, Yinhua Piao, Yunhak Oh, Woo Youn Kim, Chanyoung Park, **Sungsoo Ahn**, and Junhyeok Jeon, Progressive Multi-Agent Reasoning for Biological Perturbation Prediction, *arXiv*, 2026. [arXiv] [code]

7. Nayoung Kim, Honghui Kim, Sihyun Yu, Minkyu Kim, Seongsu Kim, and **Sungsoo Ahn**, AtomMOF: All-Atom Flow Matching for MOF-Adsorbate Structure Prediction, *arXiv*, 2026. [arXiv] [code]

8. Yunhui Jang, Seonghyun Park, Jaehyung Kim, and **Sungsoo Ahn**, INDIBATOR: Diverse and Fact-Grounded Individuality for Multi-Agent Debate in Molecular Discovery, *arXiv*, 2026. [arXiv] [code] [project]

9. Hyosoon Jang, Hyunjin Seo, Honghui Kim, Seonghyun Park, Taewon Kim, Yunhui Jang, and **Sungsoo Ahn**, A Systematic Evaluation of Co-folding Model Representations for Small-Molecule Learning, *arXiv*, 2026. [arXiv] [code]

10. Hyosoon Jang, Yunhui Jang, Jaehyung Kim, and **Sungsoo Ahn**, Can LLMs Generate Diverse Molecules? Towards Alignment with Structural Diversity, *arXiv*, 2024. [arXiv]

11. Dongyeop Woo and **Sungsoo Ahn**, Iterated Energy-based Flow Matching for Sampling from Boltzmann Densities, *arXiv*, 2024. [arXiv]

Service

1. *Track Chair (AI Scientist / Agent Systems)*, AI for Science Conference (AI4Sci Korea), Seoul, South Korea, 2026.
2. *Host*, Korea AI4Science Gathering @ ICML, Seoul, South Korea, 2026.
3. *Local Chair*, Symposium on Probabilistic Machine Learning (ProbML), Seoul, South Korea, 2026.
4. *Organizer*, AI for Science: AI Scientists—Tools, Co-authors, or Founders?, International Conference on Machine Learning (ICML), 2026.
5. *Mentor*, AI Co-Research & Education for Innovative Drug (AI-CRED) Institute, InnoCORE Program, 2025 – Present.
6. *Board of Directors*, Korean Society of AI Frontier (KSAIF), 2025 – Present.
7. *Area Chair*, International Conference on Machine Learning (ICML), 2022 – 2026.
8. *Area Chair*, International Conference on Learning Representations (ICLR), 2024 – 2026.
9. *Area Chair*, Conference on Neural Information Processing Systems (NeurIPS), 2022 – 2026.
10. *Area Chair*, International Conference on Artificial Intelligence and Statistics (AISTATS), 2022 – 2024.
11. *Area Chair*, AAAI Conference on Artificial Intelligence (AAAI), 2024.
12. *Action Editor*, Transactions of Machine Learning Research (TMLR), 2024 – Present.
13. *Workflow Chair*, International Conference on Machine Learning (ICML), 2022.
14. *Reviewer*, NeurIPS, 2018 – 2021.
15. *Reviewer*, ICML, 2019 – 2021.
16. *Reviewer*, ICLR, 2019 – 2024.
17. *Reviewer*, AISTATS, 2019 – 2021.

Invited Talks and Seminars

1. *비정형 데이터 기반 딥러닝의 과학기술 활용 및 혁신 사례*, KAIST Chief AI Officer Program, Seoul, South Korea, Nov. 2026.
2. *From Atomistic Generative Models to Amortized Free-Energy Estimation*, Generative Models and Uncertainty Quantification (GenU), Copenhagen, Denmark, Sep. 2026.
3. *Amortizing Molecular Simulation with Neural Networks*, DTU Energy, Kongens Lyngby, Denmark, Sep. 2026.
4. *AI for Molecular and Materials Discovery*, KAIST College of AI Colloquium, Daejeon, South Korea, Sep. 2026.
5. *Recent Trends in AI for Science*, Korea Supercomputing Conference (KSC), Seoul, South Korea, Sep. 2026.
6. *From Atomistic Generative Models to Amortized Free-Energy Estimation*, KAIST–Mila Prefrontal AI Workshop, Daejeon, South Korea, Aug. 2026.
7. *K-Fold: Fast Biomolecular Complex Prediction without Evolutionary Signals*, Korean Society of Artificial Intelligence in Medicine (KoSAIM), South Korea, Jul. 2026.
8. *Machine Learning Hamiltonians*, 22nd KIAS Workshop on Electronic Structure Calculations, Seoul, South Korea, Jul. 2026.
9. *Generative Approaches to Protein Conformational Landscapes*, 2026 Korean Biomolecular Science Joint Conference, South Korea, Jun. 2026.
10. *Generative Approaches to Protein Conformational Landscapes*, Korea Institute for Advanced Study (KIAS), Seoul, South Korea, Jun. 2026.
11. *비정형 데이터 기반 딥러닝의 과학기술 활용 및 혁신 사례*, KAIST Chief AI Officer Program, Seoul, South Korea, May 2026.
12. *AI Methods for Generating Scientific Hypotheses*, Samsung Advanced Institute of Technology Scientific AI Workshop, Suwon, South Korea, May 2026.
13. *Generative Models for Molecular Discovery*, Korea Institute of Science and Technology Information (KISTI), Daejeon, South Korea, Mar. 2026.
14. *비정형 데이터 기반 딥러닝의 과학기술 활용 및 혁신 사례*, KAIST Chief AI Officer Program, Seoul, South Korea, Nov. 2025.
15. *Foundation Models for Material Discovery*, Korea Institute of Materials Science (KIMS), Changwon, South Korea, Jul. 2025.
16. *Generative Models for Molecular Discovery*, Seoul National University, Seoul, South Korea, Jul. 2025.
17. *A Generative Model for Metal-Organic Frameworks*, KAIST–Mila Annual Workshop, Montreal, Canada, Jul. 2025.

18. *Machine Learning for Transition Path Sampling*, The Korea in Silico BioDesign and Discovery Society (KIDDS), Daejeon, South Korea, Jul. 2025.
19. *그래프신경망의 과학기술 활용 및 혁신 사례*, KAIST Chief AI Officer Program, Seoul, South Korea, May 2025.
20. *Generative Models for Molecular Discovery*, LG AI Research, Seoul, South Korea, May 2025.
21. *Optimization Methods for Materials Science*, POSCO Holdings, Seoul, South Korea, Dec. 2024.
22. *Structure- and Energy-Based Machine Learning for Molecules*, KAIST–Mila Prefrontal AI Workshop, Montreal, Canada, Dec. 2024.
23. *Amortizing Intractable Inference for Molecular Discovery*, Physics-Informed Machine Learning Workshop, New Mexico, United States, Nov. 2024.
24. *Opportunities and Challenges in Deep Graph Generative Models*, AI Korea, Seoul, South Korea, Nov. 2023.
25. *Machine Learning for Drug Discovery*, KAIST, Daejeon, South Korea, Oct. 2023.
26. *Graph Structured Prediction and Graph Generation with Deep Learning*, KAIST, Daejeon, South Korea, Jun. 2023.
27. *Deep Neural Networks for Graph Optimization*, University of Arizona, Tucson, Arizona, United States, Nov. 2022.
28. *Geometric Deep Learning for Drug Discovery*, Korean Artificial Intelligence Association (KAIA), Jeju, South Korea, Aug. 2022.
29. *Machine Learning for Drug Discovery*, Centre for Frontier AI Research, Singapore, Jun. 2022.
30. *Elimination Techniques in Probabilistic Graphical Models*, Skolkovo Institute of Science and Technology (Skoltech), Moscow, Russia, Jun. 2021.
31. *Guiding Deep Molecular Optimization with Genetic Exploration*, Seminar at Samsung Advanced Institute of Technology, Suwon, South Korea, Dec. 2020.
32. *Scaling Deep Reinforcement Learning to Large Combinatorial Problems*, AI & CSE Seminar, POSTECH, Pohang, South Korea, Feb. 2020.
33. *Bucket Renormalization for Approximate Inference*, Robotics Seminar, University of Oxford, Oxford, United Kingdom, Jul. 2018.
34. *Mini-Bucket Renormalization*, Physics-Informed Machine Learning Workshop, Santa Fe, New Mexico, United States, Feb. 2018.
35. *Gauge Transformation of Graphical Models*, CNLS Student Seminar, Los Alamos National Laboratory, Los Alamos, New Mexico, United States, Jul. 2017.
36. *Optimizing Gauge Transformation for Inference in Graphical Models*, Banff Workshop on Optimization and Inference for Physical Flows on Networks, Banff, Alberta, Canada, Feb. 2017.

37. *Synthesis of MCMC and Belief Propagation*, CNLS Student Seminar, Los Alamos National Laboratory, Los Alamos, New Mexico, United States, Jul. 2016.
38. *Minimum Weight Perfect Matching via Blossom Belief Propagation*, Discrete Math Seminar, KAIST, Daejeon, South Korea, Nov. 2015.

Teaching

1. *Proteins and AI (CoE499)*, KAIST, Spring 2026.
2. *Machine Learning for Molecules (AI899)*, KAIST, Fall 2025.
3. *Geometric Deep Learning (AI810)*, KAIST, Spring 2025.
4. *Introduction to Artificial Intelligence (CSED105)*, POSTECH, Fall 2023 and Fall 2024.
5. *Machine Learning for Graphs (CSED/AIGS703I)*, POSTECH, Spring 2024.
6. *Probabilistic Graphical Models (CSED/AIGS524)*, POSTECH, Spring 2022 and Spring 2023.
7. *Introduction to Machine Learning (CSED490B)*, POSTECH, Fall 2022.

Awards

1. *Outstanding Teaching Award*, KAIST, 2026.
2. *Q-Day Faculty Special Award, Research Division*, KAIST, 2025.